WORKED SOLUTION

STEP III 1987, Question 8

Pure • Parallelogram Area Inequalities

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q8

The key idea

The diagonals of a parallelogram bisect each other, so they cut it into four triangles whose areas we can write down at once using $\frac{1}{2}ab\sin C$. That gives a formula for the area in terms of the diagonals and the angle between them, and both opening results drop out of it. The set A is where two strips cross, so it is a parallelogram. If we work out its area, the second result tells us its longer diagonal must be long, and the two ends of that diagonal are the points we want.

Solution

Area in terms of the diagonals

Let the diagonals have lengths p and q and meet at an angle θ, where $0 < \theta < \pi$. The diagonals of a parallelogram bisect each other, so they split it into four triangles, each with two sides $\frac{1}{2}p$ and $\frac{1}{2}q$ meeting at the centre. Two of these triangles have angle θ there and the other two have angle $\pi - \theta$. Since $\sin(\pi - \theta) = \sin\theta$, each triangle has area $\frac{1}{2} \cdot \frac{p}{2} \cdot \frac{q}{2}\sin\theta$, so

$$\text{area} = 4 \cdot \tfrac{1}{8}pq\sin\theta = \tfrac{1}{2}pq\sin\theta.$$

If p and q are fixed, this is largest when $\sin\theta$ is largest, which is when $\theta = \frac{\pi}{2}$. So the area is maximal exactly when the diagonals are perpendicular, and the maximum area is $\frac{1}{2}pq$.

The bound by the longer diagonal

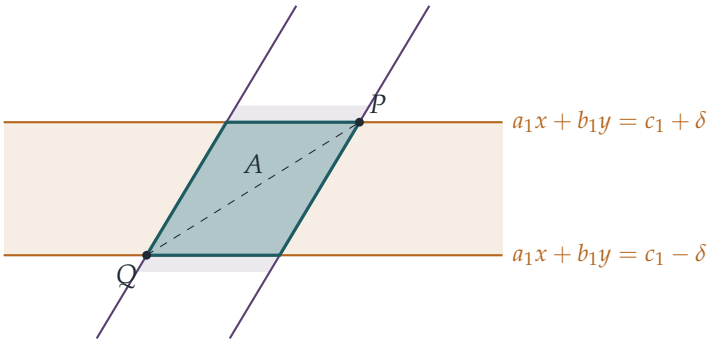
Suppose $p \geq q$, so that p is the longer diagonal (or one of two equal ones). Using $\sin\theta \leq 1$ and then $q \leq p$,

$$\text{area} = \tfrac{1}{2}pq\sin\theta \leq \tfrac{1}{2}pq \leq \tfrac{1}{2}p^2.$$

So the area is at most half the square of the longer diagonal. Equality needs $\sin\theta = 1$ and $q = p$, so it holds only for a square.

The set A is a parallelogram

Assume $\delta > 0$. (If $\delta = 0$ the bound is 0, and any single point of A will do twice.) The inequality $|a_1x + b_1y - c_1| \leq \delta$ says that $c_1 - \delta \leq a_1x + b_1y \leq c_1 + \delta$. So it describes the closed strip between the two parallel lines $a_1x + b_1y = c_1 \pm \delta$. The second inequality gives a second strip in the same way. The condition $a_1b_2 \neq a_2b_1$ says that the normal vectors $\mathbf{n}_1 = (a_1, b_1)$ and $\mathbf{n}_2 = (a_2, b_2)$ are not parallel. So the two strips cross, and neither $\mathbf{n}_1$ nor $\mathbf{n}_2$ is zero. Their intersection A is a parallelogram, bounded by two pairs of parallel lines.



The figure shows the first strip (amber) and part of the second strip (plum). Their intersection A (teal) is a parallelogram, and the dashed line is its longer diagonal.

The area of A

The distance between the parallel lines $a_1x + b_1y = c_1 \pm \delta$ is

$$h_1 = \frac{2\delta}{\sqrt{a_1^2 + b_1^2}},$$

using the formula for the distance from a point to a line. Similarly, the second strip has width $h_2 = 2\delta/\sqrt{a_2^2 + b_2^2}$. Let φ be the angle between the two families of lines, with $0 < \varphi < \pi$. The sides of A that lie along the second strip's edges cross the first strip at angle φ, so each has length $h_1 / \sin\varphi$. The perpendicular distance between these two sides is h_2, so

$$\text{area of } A = \frac{h_1 h_2}{\sin\varphi}.$$

Ask yourself How do we find $\sin\varphi$ from the coefficients?

The angle between two lines equals the angle between their normals, or its supplement, and both have the same sine. For the vectors $\mathbf{n}_1 = (a_1, b_1)$ and $\mathbf{n}_2 = (a_2, b_2)$, the area of the parallelogram they span is $|\mathbf{n}_1||\mathbf{n}_2| \sin\varphi$. That area is also $|a_1b_2 - a_2b_1|$, the size of

the 2×2 determinant. Hence

$$\sin\varphi = \frac{|a_1b_2 - a_2b_1|}{\sqrt{a_1^2+b_1^2}\sqrt{a_2^2+b_2^2}}.$$

Substituting, the square roots cancel:

$$\text{area of } A = \frac{2\delta}{\sqrt{a_1^2+b_1^2}} \cdot \frac{2\delta}{\sqrt{a_2^2+b_2^2}} \cdot \frac{\sqrt{a_1^2+b_1^2}\sqrt{a_2^2+b_2^2}}{|a_1b_2 - a_2b_1|} = \frac{4\delta^2}{|a_1b_2 - a_2b_1|}.$$

The two points

Let PQ be the longer diagonal of A, and write $P = (x_1, y_1)$ and $Q = (x_2, y_2)$. These are vertices of A, so they lie in A, because the strips include their boundary lines. By the bound from the first part,

$$\frac{4\delta^2}{|a_1b_2 - a_2b_1|} = \text{area of } A \leq \tfrac{1}{2}PQ^2,$$

so

$$(x_1 - x_2)^2 + (y_1 - y_2)^2 = PQ^2 \geq \frac{8\delta^2}{|a_1b_2 - a_2b_1|},$$

as required.

Checks

- Take $a_1 = b_2 = 1$ and $a_2 = b_1 = 0$, so that A is the square $|x - c_1| \leq \delta$, $|y - c_2| \leq \delta$. Its diagonal squared is $(2\delta)^2 + (2\delta)^2 = 8\delta^2$, which is exactly the bound. That fits, because a square is the equality case of the first part.
- Scaling: if we double a_1, b_1 and c_1, the first strip halves in width, the area of A halves, and $|a_1b_2 - a_2b_1|$ doubles. The formula behaves correctly.
- By computer we tested 10^5 random cases. We found the vertices of A by solving the boundary equations, and the shoelace area matched $4\delta^2/|a_1b_2 - a_2b_1|$ every time. The smallest ratio of (longer diagonal)^2 to the bound was 1.0007, which is never below 1.

Common slips

- Taking the strip widths to be 2δ. The width of $|a_1x + b_1y - c_1| \leq \delta$ is $2\delta/\sqrt{a_1^2+b_1^2}$.
- Forgetting to say why the chosen points lie in A. They do because they are vertices, and A includes its boundary.
- Writing the area of A as h_1h_2, which is only right when the strips are perpendicular.

Beyond STEP

The area calculation is a special case of a fundamental fact: a linear map with matrix M multiplies all areas by $|\det M|$. Here the map $(x, y) \mapsto (a_1x + b_1y,\ a_2x + b_2y)$ sends A to a square of area $4\delta^2$, so A has area $4\delta^2/|\det M|$. This is the Jacobian factor you meet when changing variables in double integrals. Strang's *Introduction to Linear Algebra* develops determinants as areas and volumes in this way.

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